

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA

Expected May 2028

Expected May 2027

Relevant Coursework: Introduction to Algorithms, Introduction to Computer Systems, Automata, Computability & Complexity, Programming Languages & Techniques, Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Made the company's AI agent usable by studios outside it** — a typed **SDK** with versioned APIs, per-session isolation, capability enforcement and **1,301 tests**, plus hosted voice and a mode-authoring Claude skill — proven inside a mode an **external builder authored** with no access to our repository or internal docs
- **Made the agent behave appropriately for the children using it** — combat targeting keyed on a hostility flag alone, so it picked fights with bosses far above their level; corrected additively, and extended its perception to a mode's custom content
- **Redirected the team's roadmap** with a **Python multi-agent tool on the Claude Agent SDK** that mined production data through read-only MCP — it showed a reported retention gain was a measurement artifact rather than a real effect, and surfaced the churn drivers that actually mattered for the next planning cycle
- **Built the company's lifecycle notification system, then made it autonomous** — it proposes sends, clears its own safety guardrails and delivers unattended at **91% device reach**, producing its first paid conversion

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Improved UI/UX and core gameplay stability across **50+ live Unity titles** in a **420M+ download** portfolio
- Shipped fixes for **100+ bugs** raised by QA and player reports, correcting Unity prefabs and C# scripts in production
- Kept all 50+ titles live on the app stores by upgrading SDKs and platform modules as requirements changed

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a team of **15+** designers and developers shipping new products for real clients, from first concept to launch
- Defined the end-to-end user flows and reusable interface systems the team shipped across multiple client deliverables

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **A Minecraft-style 3D game engine written from scratch in C++ and OpenGL** (team of 3) — I owned the world generation: **7 procedural biomes** blended from layered noise, and caves generated as you explore
- **Owned the rendering pipeline that makes the world look real** — **13 GLSL shaders** for shadows, reflective water, ambient shading and a day-night cycle, plus post-process overlays for underwater and lava

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **A 3D modeling tool in the spirit of Maya, in C++ and Qt** — edit any model's geometry directly, with a half-edge structure tracking how faces, edges and vertices connect, each selectable from a clickable Qt list
- **Smooths rough models into refined surfaces** with Catmull–Clark subdivision, the algorithm film and game studios use

CareerOS | Node.js, TypeScript, GitHub Actions

May 2026

- **A job-search platform I built for myself and run daily** — it checks **191 company career sites** every six hours and ranks **11,000+ open roles** down to the few worth reading, sorted by how well each fits what I am looking for
- **Runs unattended without losing jobs quietly** — a site that breaks or returns nothing is flagged, not silently dropped, and a daily check re-scans every site into a tracker I read each morning

TECHNICAL SKILLS

AI & Agents: Claude Agent SDK, MCP servers, Claude Code, Codex, LLM APIs (Anthropic/Gemini), agent orchestration

Languages: Python, TypeScript, JavaScript, Java, Kotlin, C++, C#, SQL, HTML/CSS

Frameworks & Infrastructure: React, Next.js, Node.js, React Native (Expo), PostgreSQL/Supabase, Deno edge functions, FCM, Docker, GCP (Cloud Run), AWS, MongoDB, Jest, PostHog, Sentry, Figma, GitHub Actions

Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing